



# ENG103: Research Writing Techniques

Participant Oriented Research, Samples, and Surveys

# Lesson outline

Participant Oriented Research

Representative Samples & Sampling Errors Surveys

Questionnaires

Question types

Questionnaire Design



# Participant Oriented Research

Participant-oriented research is the gathering of data from people rather than the things they produce or the things that affect them.

In addition, participant-oriented research can take the form of quantitative research (e.g., large scale structured surveys) or qualitative research (e.g., unstructured interviews).

Moreover, participant-oriented research can be used for *inductive research* as a means to produce hypotheses or it can be used for *deductive research* to test existing hypotheses.

# Representative Samples & Sampling Errors

In everyday life we rely on samples to make decisions about aggregates.

Similarly, when we need to reach decisions about a research population, we can conduct participant-oriented research through which we only need to test a sample of participants.

Ruane (2005) suggests that “samples, then, offer a practical solution to the daunting task of studying entire populations. We can use samples to “stand in” for a larger population. In this sense, samples can be very efficient devices – they allow us to look at the “few” in order to know about the many” (p. 105).

However, sample groups that are not representative of the population under investigation can lead to sampling errors. Leavy (2017) suggests that “in survey research it is important to consider sampling error when you draw your sample. Sampling error occurs when you have a biased sample” (p. 110).



# Representative Samples & Sampling Errors

Individuals in populations are heterogeneous (not the same as each other) in multiple ways. **What ways do you think people are different from each other in a population?**

It is very rare to find individuals in populations that are homogeneous (the same as each other). **Why do you think homogeneous populations are rare?**

Thus, **a key characteristic of a sample is that it is representative of the population.** In short, the sample should be a scaled down version of the population.

A way to ensure the sample will be representative of the population is to establish a sampling frame before forming the sample. Essentially, a sample frame is the collection of detailed information about the research population. This information can be then used to ensure the sample is representative of the population. **For example, if you were conducting research on a sample to reach a decision about the population of Prince Sultan University's students what information would we need to know about the student population?**

Regarding the importance of a sampling frame Ruane (2005) suggests that

A sampling frame refers to an exhaustive listing of all the elements that make up a research population. The sampling frame is an essential ingredient for probability sampling. Indeed, it is the sampling frame that enables the researcher to "know" the probability of elements being selected for a sample. If a researcher knows that a sampling frame contains 1,000 elements, then the researcher also knows that any one element has one chance in 1,000 of being selected for the sample. Without this total listing of all the population elements, probability sampling is impossible (p. 110).



# Representative Samples & Sampling Errors

Samples that are not representative of the population can lead to sampling errors. Leavy (2017) suggests that “in survey research it is important to consider sampling error when you draw your sample. Sampling error occurs when you have a biased sample” (p. 110).

## **A famous example of a sampling error is in the predictions surrounding the 1936 US presidential election.**

- 1936 elections in U.S.: Franklin D. Roosevelt vs. Alf Landon
- The magazine, Literary Digest, sent out about 10 million questionnaires to a list of subscribers to the magazine, and of people with telephones, cars, and various club memberships.
- About 25 % responded: Sample size of 2,4 million
- The magazine predicted Landon victory with 57 % of the vote. Actual result: Roosevelt: 61 %, Landon: 37 %.
- George Gallup's American Institute of Public Opinion predicted Roosevelt vote at 56 % of the vote with a sample size of 50,000.
- Literary Digest went bankrupt in a couple of years.

## **Why do you think Literary Digest's results were so wrong despite the large sample size?**

# Surveys

Surveys are a method to collect and analyze standardized information from participants. **Leavy (2017) explains that:**

Surveys rely on asking people standardized questions that can be analyzed statistically. They allow researchers to collect a breadth of data from large samples and generalize to the larger population from which the sample was drawn (p. 101).

**There are two main types of survey design, which are cross-sectional and longitudinal**

**Cross-sectional:** “Cross-sectional designs seek information from a sample at one point in time” (Leavy, 2017, p.101). For example, a survey to collect and analyze data in relation to how a population rate the taste of a new diet food product.

**Longitudinal:** “Longitudinal designs occur at multiple times in order to measure change over time” (Leavy, 2017, p. 101). For example, a survey to collect and analyze data in relation to how a new diet food product impacts people's health over a ten-year period.



# Questionnaires

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**As mentioned, surveys involve the collection and analysis of data, but a questionnaire is limited just to the collection phase of data.** For example, Leavy (2017) explains that “questionnaires are the primary data collection tool in survey research. A questionnaire is also referred to as the survey instrument” (p.101). Also, Matthews and Ross (2010) define a questionnaire as follows:

(1) a list of questions each with a range of answers; (2) a format that enables standardized, relatively structured, data to be gathered about each of a (usually) large number of cases (cited in Saldanha and O'Brien 2014 pg. 151).

## **A questionnaire has several advantages over interviews:**

A questionnaire can reach many participants

The data is easy to collate, especially with forced-choice questions

The data results are, theoretically, easy to reproduce

## **However, a questionnaire has several inherent disadvantages:**

Constrained responses (sometimes forced-choice questions do not tell the full story)

Finding appropriate participants

Low response rate



# Questionnaires

## Questionnaire design

**Before drafting questions, it is necessary to clearly identify the objective of the research and how questions can contribute to achieve the research objective.**

Leavy (2017) explains that “here must be clear and justifiable links between your indicators (questions) and the concepts you say you are measuring in order to produce a valid instrument” (p. 101)

Leavy (2017) gives the example of research to determine if there is a link between students' satisfaction with their college roommate and their overall satisfaction with their college experience. However, as Leavy notes these are both multidimensional concepts.

**For example, what would be the variables of satisfaction with a college roommate? That is to say, what are factors you would measure satisfaction with a college roommate?**

Leavy (2017) explains that “let's say you come up with the following three dimensions of “cleanliness”: personal items kept orderly, communal items kept orderly, and personal hygiene. You would then come up with several questions to measure each of those dimensions of “cleanliness.” All of those questions together measure the concept of “cleanliness.”” (pp. 102-103.)

Moreover, Leavy (2017) asserts that “the more multidimensional a variable concept is, the more questions you will likely ask in regard to that specific variable” (Leavy, 2017, p. 103).

# Questionnaires

**TABLE 4.5. Example of a Two-Column Approach**

| Variable: Cleanliness        | Draft questions (in idea form)                                                                                                                                         |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal items kept orderly. | Roommate regularly makes his/her bed.<br>Roommate puts away his/her clothes in drawers/closet.<br>Roommate keep personal items on his/her side of the room.            |
| Communal items kept orderly. | Roommate hogs the common-use items in the room.<br>Roommate returns common-use items to designated places.<br>Roommate cleans common-use items if he/she dirties them. |
| Personal hygiene.            | Roommate has good personal hygiene habits.<br>Roommate cleans bedding regularly.<br>Roommate wears clean clothing.                                                     |

Leavy, 2017, p. 103.



# Question types

**There are two main types of questions that can be used in a survey:**

1) Open-ended questions: the participant is free to answer the question in anyway. The advantage of this approach is that it can yield unexpected responses. The disadvantage is that the answers are difficult to process in a quantitative study as the data will not be structured.

2) forced-choice: the participant is forced to answer the question in accordance with a limited number of potential answers. The advantage of this approach is that it can yield structured information that is easy to process in a quantitative study. The disadvantage is that it may miss unexpected responses (unknown unknowns).

Regarding forced-choice Leavy (2017) suggests that “different types of forced-choice questions include, but are not limited to, multiple choice, dichotomous, checklists, and scales (rating scale, Likert scale) [...]” (p. 104)."

# Question types

**Multiple choice:** A question with several response options (typically four to five) is provided, and respondents select a single response.

When are you most likely to confide something personal to your roommate?

- ☐ When I have a problem and need advice
- ☐ When something good has happened
- ☐ When I have a secret I want to share
- ☐ Never

**Dichotomous:** Provide a statement with two response options (such as yes/no or true/false).

I talk to my roommate about personal matters.

- ☐ True
- ☐ False

**Checklist:** Provide a question with several response options and direct respondents to check all that apply.

I talk to my roommate about personal matters when (please select *all* that apply) . . .

- ☐ I need to confide in someone about a secret
- ☐ I need to seek advice about a problem
- ☐ I want to share good news
- ☐ Never

Leavy, 2017, p. 104.



# Question types

**Rating scale:** Provide a statement or question with response options on a continuum and instruct respondents to select a single response.

**How often do you talk to your roommate about personal matters?**

- ☐ Very frequently
- ☐ Somewhat frequently
- ☐ Occasionally
- ☐ Rarely
- ☐ Never

**Likert scale:** Provide a statement with responses that indicate level of agreement and ask respondents to select a single response. There are usually four or five response choices, depending on whether you include a neutral, midpoint option or force a non-neutral choice.

**When I have a problem, I greatly value the advice of my roommate.**

- ☐ Strongly agree
- ☐ Somewhat agree
- ☐ Neutral
- ☐ Somewhat disagree
- ☐ Strongly disagree

Leavy, 2017, p. 105.

# Question types

**Also, in regard to a Likert Scale Saldanha and O'Brien (2014) suggest that:**

A very common device in questionnaires is the 'Likert scale', which was developed by the American psychologist Rensis Likert. A Likert scale commonly offers a series of five, seven or nine responses along a continuum of 'strongly agree to strongly disagree'. Some researchers favour an even-numbered scale which, they believe, prevents respondents from conveniently selecting the mid-point on the scale (usually 'neither agree nor disagree' or 'not sure') (pp. 157-58).



# Response choices

**When formulating response choices in a questionnaire it is important that they are distinct from each other and thus do not overlap.**

For example, Leavy (2017) argues that:

Response choices must be mutually exclusive and exhaustive. Mutually exclusive means that there is no overlap in response items, and exhaustive means that all of the possible responses a respondent might wish to select are available (p. 106).

# Response choices

**My roommate makes his/her bed:**

- ☐ 1–2 times per week
  - ☐ 2–3 times per week
  - ☐ 4–5 times per week
  - ☐ 6–7 times per week
  - ☐ Never
- } Not mutually exclusive

Note: The above answers are *not* mutually exclusive. If the roommate makes the bed twice a week, there are two possible response options when there should only be one.

**My roommate makes his/her bed:**

- ☐ 1–2 times per week
  - ☐ 3–4 times per week
  - ☐ 5–6 times per week
  - ☐ 7 days per week
- } Not exhaustive

Note: The above answers are *not* exhaustive. If the roommate never makes his/her bed, there is no accurate option for the respondent to select.

Now let's look at a final example in which the response options are both mutually exclusive and exhaustive.

**My roommate makes his/her bed:**

- ☐ 1–2 times per week
- ☐ 3–4 times per week
- ☐ 5–6 times per week
- ☐ 7 days per week
- ☐ Never

Leavy, 2017, p. 106.



# Questionnaire Design

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**The first step** in designing a survey is to be clear on the goals of the research.

**The second step** is to formulate questions that are beneficial to achieving those research goals. Moreover, the questions should not be excessive.

**A wrongly designed questions can jeopardize the entire research project. Therefore, the following points should be adhered to when formulating a questionnaire:**

- The avoidance of controversial questions, which may jar with the participant.
- The avoidance of ambiguous language.
- The use of neutral wording so not to prompt a response.

**Also, before sending your questionnaire to your participants it is important to undertake preliminary research, which is a way to test your questionnaire and identify any issues. This can be undertaken by sending your questionnaire to:**

- Fellow academics
- A sample group of participants
- An ethics committee

**The responses should be used to improve your questionnaire design.**

# Questionnaire Distribution

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**The participants' answers to the questionnaire can be sought through various means, but the main methods are:**

- ▶ the researcher asking participants to complete the questionnaire in their presence.
  - ▶ An organized event, which participants attend
  - ▶ The direct canvassing of participants without prior notification
- ▶ the researcher asking participants to complete the questionnaire in their absence.
  - ▶ Using online means such as Google forms
  - ▶ Emailing the questionnaire to participants



# References

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Leavy, P. (2017). *Research Design: Quantitative, Qualitative, Mixed Methods, Arts-Based, and Community-Based Participatory Research Approaches*. The Guilford Press.

Ruane, J. M. (2005). *Essentials of Research Methods A Guide to Social Science Research*. Blackwell Publishing.